

Hybrid Multimodal Transformer — The Recommended Recognition Core

Per-model deep dive · design, training, and why it beats every component

Result: 0.642 accuracy · 0.639 weighted F1 · WER 0.358 · fold SD ± 0.184

Two-stream feature-level fusion: ST-GCN (graph) + Transformer (attention), trained end-to-end

1 Theory and design

The hybrid combines two complementary views of a sign into one trained model. The **graph stream** (ST-GCN) encodes anatomical structure — how connected joints move together — and is the most stable deep model in the study. The **attention stream** (Transformer) encodes long-range temporal relationships — how any moment of the sign relates to any other — and is the most expressive single model. Each stream produces an embedding; the two embeddings are **concatenated (feature-level fusion)** and a fusion head classifies. Because both streams are trained together, the head learns how structure and motion *interact*, not just their separate votes.

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graph : skeleton(75 joints) → ST-GCN blocks → pooled embedding (64)
landmark: hands+pose(225) → Transformer encoder → pooled embedding (96)
fuse   : concat[64+96] → MLP → 5 classes (trained end-to-end)
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Two fusion strategies were tested — and the choice matters.

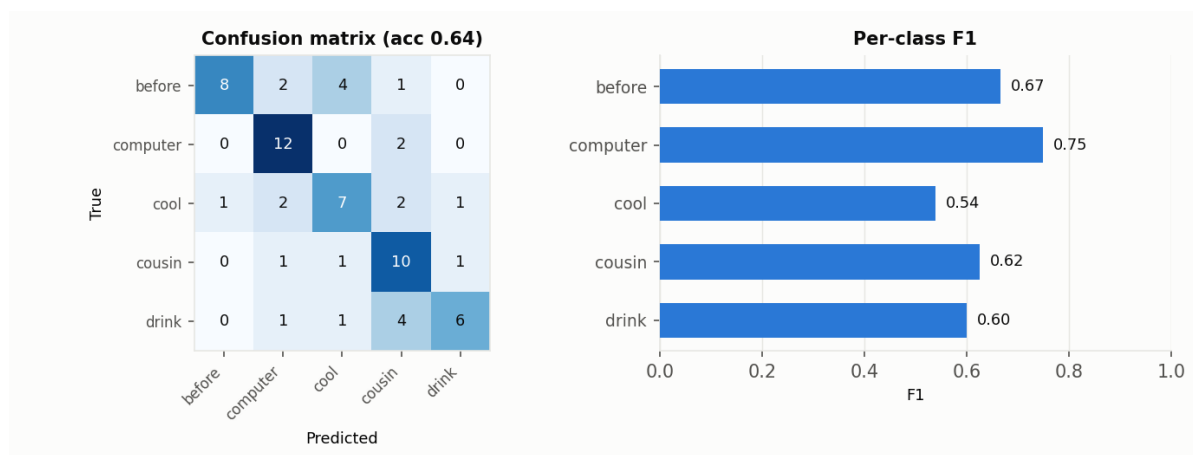
Feature-level (two-stream): fuse the streams' internal embeddings and train jointly → 0.642. **Decision-level**

(soft-vote): train BiLSTM/ST-GCN/Transformer separately and average their output probabilities → 0.478.

Feature fusion wins decisively because it can learn cross-stream interactions; soft-voting only averages three data-starved models and mostly averages their noise.

2 Results

Metric	Value (5-fold CV)
Accuracy	0.642
Weighted precision	0.679
Weighted recall	0.642
Weighted F1	0.639
Word Error Rate (WER)	0.358
Fold accuracies	0.36, 0.57, 0.69, 0.69, 0.92
Cross-val SD	± 0.184
Distinct classes predicted	5 of 5
Majority baseline	0.224



Two-stream hybrid confusion matrix and per-class $F1$ — the strongest diagonal in the study and the only model with $F1 \geq 0.54$ on every class.

3 Why it wins

The hybrid reaches 0.642 / $F1$ 0.639 — above the best single deep model (Transformer, 0.522) and the non-parametric baseline (DTW, 0.567). It inherits ST-GCN's low variance *and* the Transformer's expressiveness: the graph stream keeps the model anchored when data is scarce, while the attention stream lifts the ceiling. It is also the most **balanced** model — the only one scoring $F1 \geq 0.54$ on all five classes, where every single model has at least one class it fails badly.

4 Honest limitations

The hybrid's fold spread is the highest in the study ($SD \pm 0.184$; folds 0.36, 0.57, 0.69, 0.69, 0.92) — its lead is driven partly by one very strong fold. The direction is clear and consistent (it tops the pooled ranking), but at ~13 examples per class the *margin* is not yet statistically firm. A paired McNemar test against DTW and, above all, more data are needed before the advantage is stated as a headline number. It also currently drops the face; adding a facial-expression stream is a natural third modality.

5 Recommendation

This is the model to carry forward.

Build the recognition core of the KSL system as this **two-stream ST-GCN + Transformer hybrid**. It is the only model that empirically beats the DTW baseline, it is the most balanced across classes, and its architecture is exactly the 'Hybrid Multimodal Transformer' the thesis proposes — now with measured evidence behind it. Immediate next steps: signer-independent evaluation, a paired significance test, more data per class, and a third (facial) stream.